
Nerfstudio를 이용한 DokiDoki Minseok 만들기

1 환경 설치

```
adminuser@ubuntu:~$ docker pull ghcr.io/nerfstudio-project/nerfstudio:latest
latest: Pulling from nerfstudio-project/nerfstudio
Digest: sha256:b59b8e1012d7a43679d3234b3de9c8416a4b8435fcfbf21b9d8c4494b8563f19e
Status: Image is up to date for ghcr.io/nerfstudio-project/nerfstudio:latest
ghcr.io/nerfstudio-project/nerfstudio:latest
adminuser@ubuntu:~$ docker run -it --gpus all \
--name strong_yujin \
-p 7007:7007 \
-v $PWD:/workspace \
ghcr.io/nerfstudio-project/nerfstudio:latest \
bash
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint strong_yujin (1d3ea2e85929f0c627d284767b535601451f22d8a0af27abfb7b3e6bc6ebc27f): Bind for 0.0.0.0:7007 failed: port is already allocated

Run 'docker run --help' for more information
adminuser@ubuntu:~$ docker start -ai strong_yujin

=====
== CUDA ==
=====

CUDA Version 11.8.0

Container image Copyright (c) 2016-2023, NVIDIA CORPORATION & AFFILIATES. All rights reserved.

This container image and its contents are governed by the NVIDIA Deep Learning Container License.
By pulling and using the container, you accept the terms and conditions of this license:
https://developer.nvidia.com/ngc/nvidia-deep-learning-container-license

A copy of this license is made available in this container at /NGC-DL-CONTAINER-LICENSE for your convenience.

root@86d1982b6239:/#
```

```
adminuser@ubuntu:~$ docker run -it --gpus all \
--name strong_yujin \
--dns 8.8.8.8 \
--dns 1.1.1.1 \
-p 7008:7007 \
-v $PWD:/workspace \
ghcr.io/nerfstudio-project/nerfstudio:latest \
bash
```

```
adminuser@ubuntu:~$ docker rm -f strong_yujin
docker run -it \
--name strong_yujin \
--gpus '"device=0"' \
-p 7020:7007 \
-v $PWD:/workspace \
ghcr.io/nerfstudio-project/nerfstudio:latest \
bash
Error response from daemon: No such container: strong_yujin

=====
== CUDA ==
=====

CUDA Version 11.8.0

Container image Copyright (c) 2016-2023, NVIDIA CORPORATION & AFFILIATES. All rights reserved.

This container image and its contents are governed by the NVIDIA Deep Learning Container License.
By pulling and using the container, you accept the terms and conditions of this license:
https://developer.nvidia.com/ngc/nvidia-deep-learning-container-license

A copy of this license is made available in this container at /NGC-DL-CONTAINER-LICENSE for your convenience.
```

2 Train

```
bash: cd: data: NO SUCH FILE OR DIRECTORY
root@86e7e3bcb3af:/# ns-download-data nerfstudio --capture-name=poster
Downloading...
From (original): https://drive.google.com/uc?id=1FceQ5DX7bbTbHeL26t0x6ku56cwsRs6t
From (redirected): https://drive.google.com/uc?id=1FceQ5DX7bbTbHeL26t0x6ku56cwsRs6t&confirm=t&uu
id=d14cf618-cf41-4b58-ab9c-1b086f0d53e6
To: /data/nerfstudio/poster.zip
2%|█| 17.3M/749M [00:02<01:35, 7.64MB/s]
```

```
ns-download-data nerfstudio --capture-name=poster
```

```
ns-train nerfacto --data data/nerfstudio/poster
```

Training Finished

Config File

Checkpoint Directory

outputs/poster/nerfacto/2026-03-07_080202/config.yml

outputs/poster/nerfacto/2026-03-07_080202/nerfstudio_models

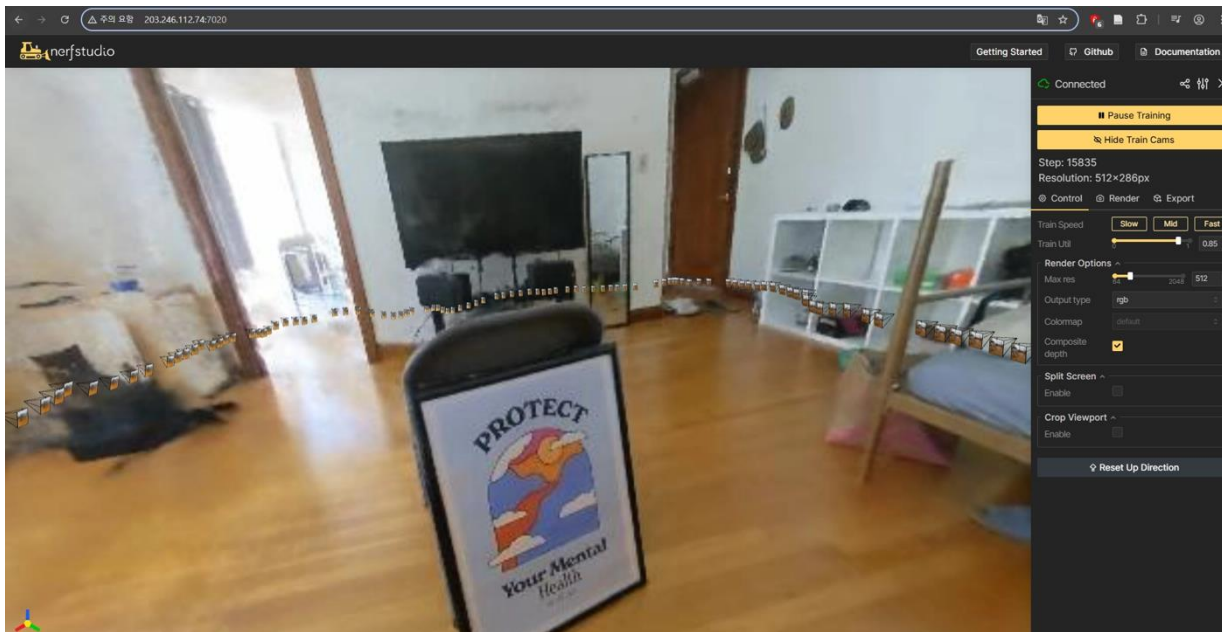
Use ctrl+c to quit

wrapping.

Step (% Done)	Train Iter (time)	ETA (time)	Train Rays / Sec	Vis Rays / Sec
15160 (50.53%)	47.076 ms	11 m, 38 s	89.42 K	
15170 (50.57%)	45.347 ms	11 m, 12 s	95.87 K	
15180 (50.60%)	41.829 ms	10 m, 19 s	104.88 K	
15190 (50.63%)	42.440 ms	10 m, 28 s	98.94 K	
15200 (50.67%)	49.774 ms	12 m, 16 s	87.71 K	
15210 (50.70%)	49.957 ms	12 m, 18 s	87.51 K	
15220 (50.73%)	49.721 ms	12 m, 14 s	83.56 K	
15230 (50.77%)	49.008 ms	12 m, 3 s	84.96 K	
15240 (50.80%)	45.865 ms	11 m, 16 s	90.11 K	
15250 (50.83%)	44.607 ms	10 m, 57 s	93.25 K	

Viewer running locally at: <http://localhost:7007> (listening on 0.0.0.0)

2 Train

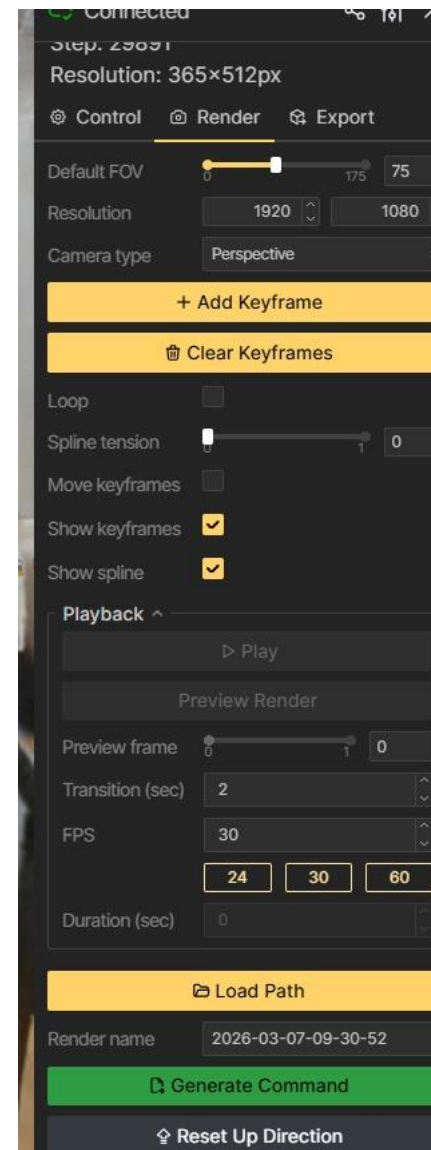


```

adminuser@ubuntu:~$ docker rm -f strong_yujin
docker run -it \
--name strong_yujin \
--gpus '"device=0"' \
-p 7020:7007 \ ...
29980 (99.93%)    44.850 ms      896.998 ms     92.11 K
29990 (99.97%)    48.769 ms      487.689 ms     86.56 K
29999 (100.00%)
-----
Viewer running locally at: http://localhost:7007 (listening on 0.0.0.0)
-----
🎉 Training Finished 🎉
-----
Config File      | outputs/poster/nerfacto/2026-03-07_093029/config.yml
Checkpoint Directory | outputs/poster/nerfacto/2026-03-07_093029/nerfstudio_models

Use ctrl+c to quit

```



Connected Step: 29999

Resolution: 365x512px

Control Render Export

Default FOV 0 175 75

Resolution 1920 1080

Camera type Perspective

+ Add Keyframe

Clear Keyframes

Loop ☐

Spline tension 0 1 0

Move keyframes ☐

Show keyframes ☒

Show spline ☒

Playback ^

Play

Preview Render

Preview frame 0 1 0

Transition (sec) 2

FPS 30

24 30 60

Duration (sec) 0

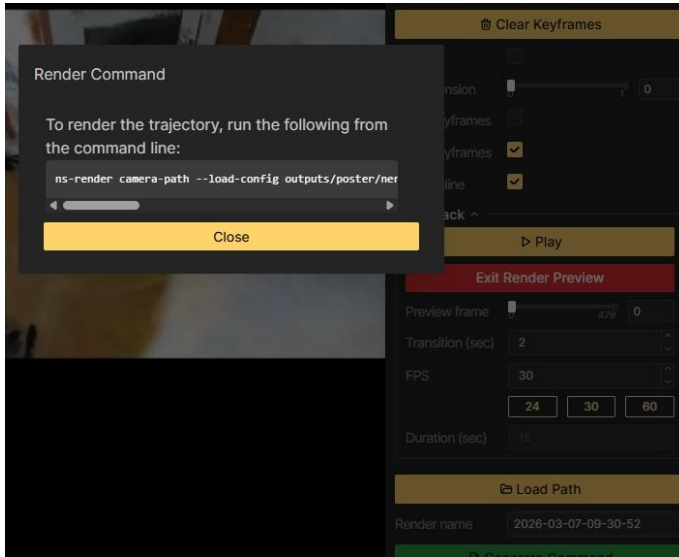
Load Path

Render name 2026-03-07-09-30-52

Generate Command

Reset Up Direction

2 Train



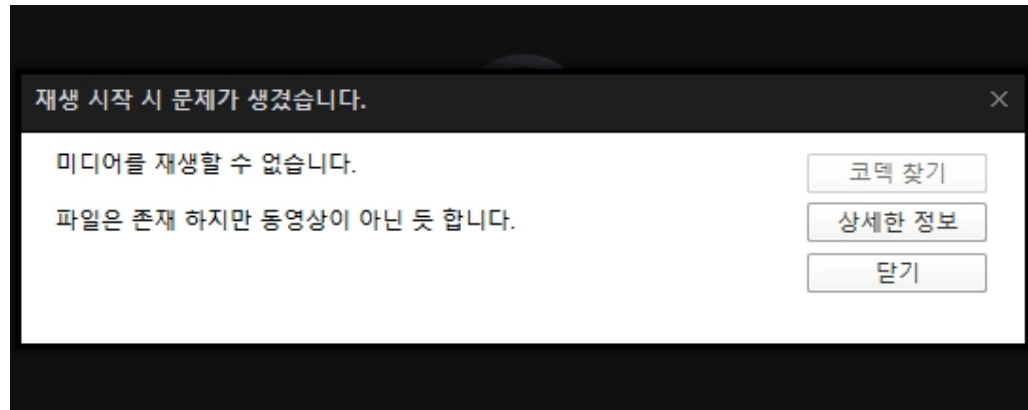
```
ns-render camera-path --load-config outputs/poster/nerfacto/2026-03-07_09302
```

사용중인 워크스페이스에 복붙

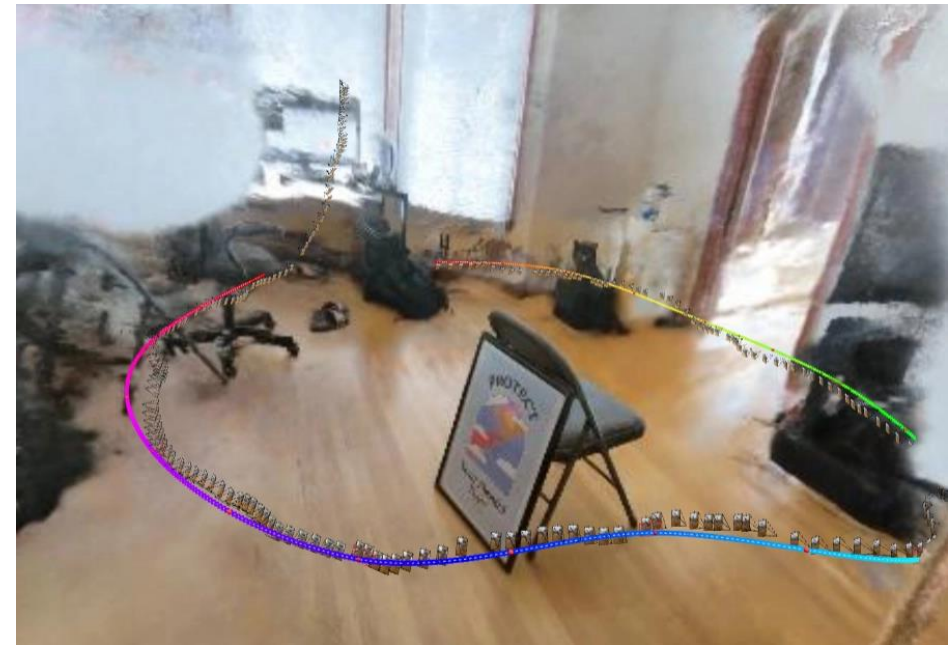
```
adminuser@ubuntu:~$ docker exec -it strong_yujin bash
to 75+.
tiny-cuda-nn warning: FullyFusedMLP is not supported for the selected architecture 70.
Falling back to CutlassMLP. For maximum performance, raise the target GPU architecture
to 75+.
tiny-cuda-nn warning: FullyFusedMLP is not supported for the selected architecture 70.
Falling back to CutlassMLP. For maximum performance, raise the target GPU architecture
to 75+.
tiny-cuda-nn warning: FullyFusedMLP is not supported for the selected architecture 70.
Falling back to CutlassMLP. For maximum performance, raise the target GPU architecture
to 75+.
tiny-cuda-nn warning: FullyFusedMLP is not supported for the selected architecture 70.
Falling back to CutlassMLP. For maximum performance, raise the target GPU architecture
to 75+.
Loading latest checkpoint from load_dir
✅ Done loading checkpoint from outputs/poster/nerfacto/2026-03-07_093029/nerfstudio_models/step-000029999.ckpt
Creating trajectory video
Rendering 1/480(0.2%) ? -:-:- 0:00:04
```

튕겨서 다시 렌더링

2 Train



문제 발생 -> 키프레임 설정을 안 해서 생김



2 Train

아 이거 상황 이해됐어.

파일은 정상인데 **Windows** 플레이어가 못 여는 케이스야. (NeRF 렌더에서 가끔 있음)

네 `ffprobe` 결과 보면 영상은 완전히 정상:

```
H264
1920x1080
30 fps
50 sec
```

그래서 문제는 **MP4 컨테이너의 moov atom** 위치일 가능성이 매우 높아.

Windows 기본 플레이어가 이거에 민감함.

해결은 **1초면 끝나**.

컨테이너에서 이거 실행해:

```
</> Bash

ffmpeg -i /renders/minseok1_processed/2026-03-08-06-11-21.mp4 \
-c copy -movflags +faststart \
/renders/minseok1_processed/2026-03-08-06-11-21_fixed.mp4
```

설명:

```
-c copy          → 재인코딩 안함 (1~2초)
-movflags +faststart → mp4 인덱스를 앞으로 이동
```

그러면 새 파일 생김:

```
2026-03-08-06-11-21_fixed.mp4
```

이거 다운받아서 열면 **100% 열릴 확률** 높아.

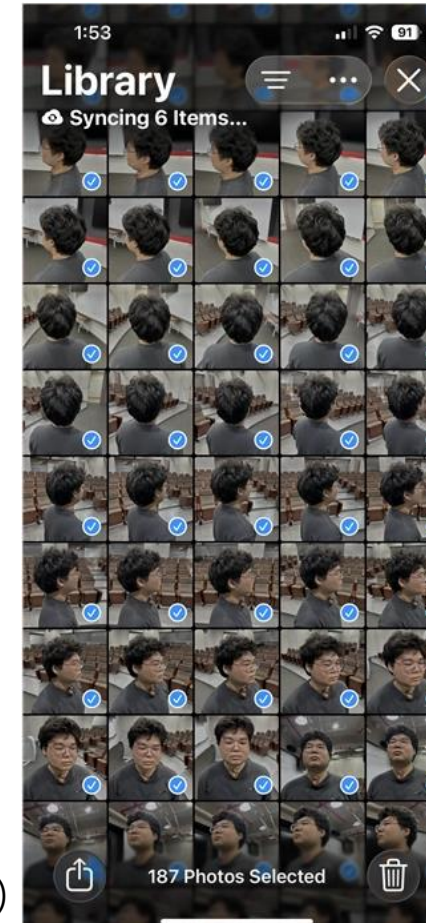
2 Train



3 DokiDoki Minseok

궁금한게 생겼습니다.

1. 사람 복원 궁금
22. 더 복잡한 배경 복원 궁금
333. 유리 같이 투명한 물체는 어떻게 복원 되는가(ex 안경)
4444. 전체가 아니라 일부만 찍는다면 어떻게 복원 되는가



아이폰 14로 DokiDoki Miunseok을 열심히 찍어줍니다 (187장)

3 DokiDoki Minseok

```
dev
root@00939cb082b8:/# cd data
root@00939cb082b8:/data# ls
nerfstudio
root@00939cb082b8:/data# cd nerfstudio/
root@00939cb082b8:/data/nerfstudio# ls
poster
root@00939cb082b8:/data/nerfstudio# mkdir minseok1
root@00939cb082b8:/data/nerfstudio# ls
minseok1 poster
root@00939cb082b8:/data/nerfstudio# cd minseok1/
root@00939cb082b8:/data/nerfstudio/minseok1# ls
root@00939cb082b8:/data/nerfstudio/minseok1# mkdir images
root@00939cb082b8:/data/nerfstudio/minseok1# ls
images
root@00939cb082b8:/data/nerfstudio/minseok1# cd images
root@00939cb082b8:/data/nerfstudio/minseok1/images#
```

폴더 만들기

```
root@00939cb082b8:/data/nerfstudio/minseok1/images# ls
IMG_2860.JPG IMG_2879.JPG IMG_2898.JPG IMG_2917.JPG IMG_2936.JPG IMG_2956.JPG IMG_2975.JPG IMG_2994.JPG IMG_3014.JPG IMG_3033.JPG
IMG_2861.JPG IMG_2880.JPG IMG_2899.JPG IMG_2918.JPG IMG_2937.JPG IMG_2957.JPG IMG_2976.JPG IMG_2995.JPG IMG_3015.JPG IMG_3034.JPG
IMG_2862.JPG IMG_2881.JPG IMG_2900.JPG IMG_2919.JPG IMG_2938.JPG IMG_2958.JPG IMG_2977.JPG IMG_2996.JPG IMG_3016.JPG IMG_3035.JPG
IMG_2863.JPG IMG_2882.JPG IMG_2901.JPG IMG_2920.JPG IMG_2939.JPG IMG_2959.JPG IMG_2978.JPG IMG_2997.JPG IMG_3017.JPG IMG_3036.JPG
IMG_2864.JPG IMG_2883.JPG IMG_2902.JPG IMG_2921.JPG IMG_2940.JPG IMG_2960.JPG IMG_2979.JPG IMG_2998.JPG IMG_3018.JPG IMG_3037.JPG
IMG_2865.JPG IMG_2884.JPG IMG_2903.JPG IMG_2922.JPG IMG_2941.JPG IMG_2961.JPG IMG_2980.JPG IMG_2999.JPG IMG_3019.JPG IMG_3038.JPG
IMG_2866.JPG IMG_2885.JPG IMG_2904.JPG IMG_2923.JPG IMG_2942.JPG IMG_2962.JPG IMG_2981.JPG IMG_3001.JPG IMG_3020.JPG IMG_3039.JPG
IMG_2867.JPG IMG_2886.JPG IMG_2905.JPG IMG_2924.JPG IMG_2943.JPG IMG_2963.JPG IMG_2982.JPG IMG_3002.JPG IMG_3021.JPG IMG_3040.JPG
IMG_2868.JPG IMG_2887.JPG IMG_2906.JPG IMG_2925.JPG IMG_2945.JPG IMG_2964.JPG IMG_2983.JPG IMG_3003.JPG IMG_3022.JPG IMG_3041.JPG
IMG_2869.JPG IMG_2888.JPG IMG_2907.JPG IMG_2926.JPG IMG_2946.JPG IMG_2965.JPG IMG_2984.JPG IMG_3004.JPG IMG_3023.JPG IMG_3042.JPG
IMG_2870.JPG IMG_2889.JPG IMG_2908.JPG IMG_2927.JPG IMG_2947.JPG IMG_2966.JPG IMG_2985.JPG IMG_3005.JPG IMG_3024.JPG IMG_3043.JPG
IMG_2871.JPG IMG_2890.JPG IMG_2909.JPG IMG_2928.JPG IMG_2948.JPG IMG_2967.JPG IMG_2986.JPG IMG_3006.JPG IMG_3025.JPG IMG_3044.JPG
IMG_2872.JPG IMG_2891.JPG IMG_2910.JPG IMG_2929.JPG IMG_2949.JPG IMG_2968.JPG IMG_2987.JPG IMG_3007.JPG IMG_3026.JPG IMG_3045.JPG
IMG_2873.JPG IMG_2892.JPG IMG_2911.JPG IMG_2930.JPG IMG_2950.JPG IMG_2969.JPG IMG_2988.JPG IMG_3008.JPG IMG_3027.JPG IMG_3046.JPG
IMG_2874.JPG IMG_2893.JPG IMG_2912.JPG IMG_2931.JPG IMG_2951.JPG IMG_2970.JPG IMG_2989.JPG IMG_3009.JPG IMG_3028.JPG IMG_3047.JPG
IMG_2875.JPG IMG_2894.JPG IMG_2913.JPG IMG_2932.JPG IMG_2952.JPG IMG_2971.JPG IMG_2990.JPG IMG_3010.JPG IMG_3029.JPG IMG_3048.JPG
IMG_2876.JPG IMG_2895.JPG IMG_2914.JPG IMG_2933.JPG IMG_2953.JPG IMG_2972.JPG IMG_2991.JPG IMG_3011.JPG IMG_3030.JPG
IMG_2877.JPG IMG_2896.JPG IMG_2915.JPG IMG_2934.JPG IMG_2954.JPG IMG_2973.JPG IMG_2992.JPG IMG_3012.JPG IMG_3031.JPG
IMG_2878.JPG IMG_2897.JPG IMG_2916.JPG IMG_2935.JPG IMG_2955.JPG IMG_2974.JPG IMG_2993.JPG IMG_3013.JPG IMG_3032.JPG
```

서버로 DokiDoki Minseok 187명 전송

```
colmap_utils.p
[START] Sun Mar 8 05:43:23 UTC 2026
[HEARTBEAT] Sun Mar 8 05:43:23 UTC 2026 ns-process-data is still running...
[HEARTBEAT] Sun Mar 8 05:43:33 UTC 2026 ns-process-data is still running...
[HEARTBEAT] Sun Mar 8 05:43:43 UTC 2026 ns-process-data is still running...
[HEARTBEAT] Sun Mar 8 06:01:43 UTC 2026 ns-process-data is still running...
[05:43:53] [HEARTBEAT] Sun Mar 8 06:01:50 UTC 2026 ns-process-data is still running...
[06:01:50] 🚀 Done COLMAP bundle adjustment.
y:348
[HEARTBEAT] y:173
[HEARTBEAT] [HEARTBEAT] Sun Mar 8 06:01:53 UTC 2026 ns-process-data is still running...
[HEARTBEAT] [HEARTBEAT] Sun Mar 8 06:02:03 UTC 2026 ns-process-data is still running...
[06:02:10] 🚀 Done refining intrinsics.
[06:02:17] 🚀 🚀 🚀 All DONE 🚀 🚀 🚀
y:135
Starting with 187 images
y:138
Colmap matched 187 images
y:138
COLMAP found poses for all images, CONGRATS!
y:138
[END] Sun Mar 8 06:02:18 UTC 2026 exit_code=0
```

```
minseok1_processed
├── colmap
├── images
├── images_2
├── images_4
├── images_8
├── sparse_pc.ply
├── transforms.json
├── minseok1_processed
│   └── colmap
│       └── sparse
│           └── 0
│               ├── cameras.bin
│               ├── images.bin
│               ├── points3D.bin
│               └── project.ini
└── database.db
```

DokiDoki Minseok
전처리 완료

DokiDoki Minseok 가공

폴더

3044.JPG
3045.JPG
3046.JPG
3047.JPG
3048.JPG

```
nohup ns-train nerfacto \
  --data /data/nerfstudio/minseok1_processed \
  --vis viewer \
  > /loglog/minseok_train_0308.log 2>&1 &
```

```
y:348 [06:01:50] 🏁 Done colmap bundle adjustment. colmap_utils.p
[HEARTBEAT] y:173
[HEARTBEAT] [HEARTBEAT] Sun Mar 8 06:01:53 UTC 2026 ns-process-data is still running...
[HEARTBEAT] [HEARTBEAT] Sun Mar 8 06:02:03 UTC 2026 ns-process-data is still running...
[05:44:26] [06:02:10] 🏁 Done refining intrinsics. colmap_utils.p
y:137 y:184
Downloading [HEARTBEAT] Sun Mar 8 06:02:13 UTC 2026 ns-process-data is still running...
[06:02:17] 🏁 🏁 🏁 All DONE 🏁 🏁 🏁 images_to_nerfstudio_dataset.p
y:135
Starting with 187 images images_to_nerfstudio_dataset.p
y:138
Colmap matched 187 images images_to_nerfstudio_dataset.p
y:138
COLMAP found poses for all images, CONGRATS! images_to_nerfstudio_dataset.p
y:138
[END] Sun Mar 8 06:02:18 UTC 2026 exit_code=0
```

```

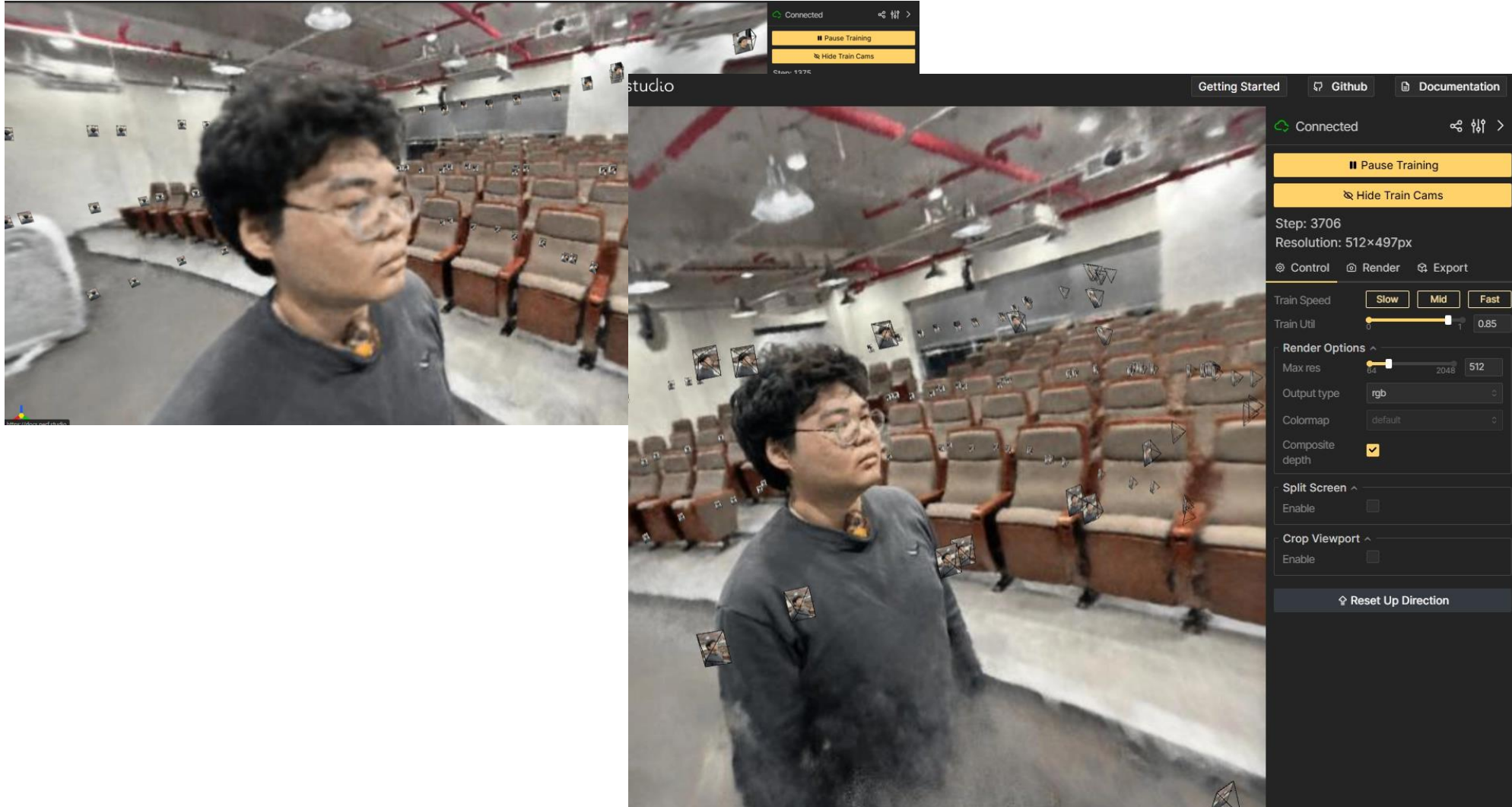
> images_8
├── sparse_pc.ply
└── transforms.json
└── minseok_i_processed
    ├── colmap
    └── sparse
        └── 0
            ├── cameras.bin
            ├── images.bin
            ├── points3D.bin
            ├── project.ini
            └── database.db

```

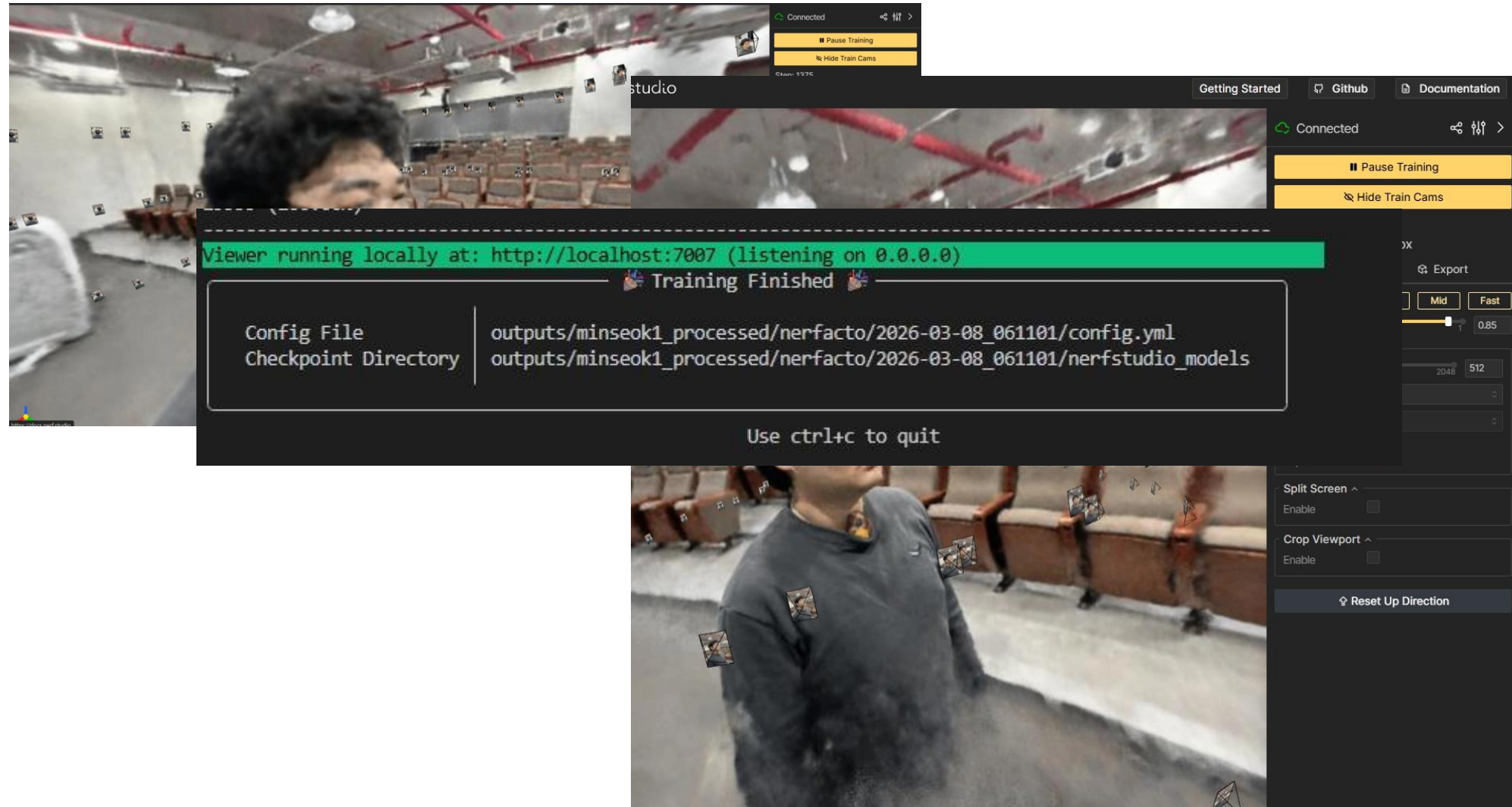
전처리 완료

데이터 가공

3 DokiDoki Minseok



3 DokiDoki Minseok



3 DokiDoki Minseok



```
nohup bash -c '  
echo "[START] $(date)"  
ns-render camera-path \  
  --load-config /outputs/minseok1_processed/nerfacto/2026-03-08_061101/confi  
  --camera-path-filename /data/nerfstudio/minseok1_processed/camera_paths/20  
  --output-path /renders/minseok1_processed/2026-03-08-06-11-21.mp4  
STATUS=$?  
echo "[END] $(date) exit_code=$STATUS"  
exit $STATUS  
' > /loglog/minseok_render_0308.log 2>&1 &
```

키프레임 찍고 렌더링 진행

3 DokiDoki Minseok

